



Risk-averse operation of energy-water nexus using information gap decision theory

Peiru Jian^a, Qun Guo^{b,*}, Sayyad Nojavan^{c,*}

^a Wuchang Shouyi University, Huadian Hubei Power Generation Co.Ltd., Wuhan, 430205, Hubei Province, China

^b School of Business Administration, Hubei University of Economics, Wuhan, 430205, Hubei Province, China

^c Department of Electrical Engineering, University of Bonab, Bonab, Iran



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ABSTRACT

Nowadays, nexus systems spread in modern societies because of their ability in incorporation of the dependencies of the interconnected resources water and energy. Other parameters such as cost-effectiveness, environmental issues can be considered as the advantages of the mentioned system. Besides the mentioned positive features, the most significant barrier, the uncertainty of the demand energy, still needs to be taken into account. So, information gap decision theory (IGDT) is applied to the proposed nexus system to analyze the water and power demand uncertainties levels. GAMS software is used to solve the proposed risk-based model. In the proposed IGDT method, risk-averse and risk-neutral strategies are applied to the proposed nexus system, and the obtained results are analyzed with each other. Obtained results indicate the robustness of studied system under uncertainties of water and power demands. For instance, $\alpha = 5.7\%$, total robustness cost of the studied nexus system is increased by approximately 5250.

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1. Introduction

Combining the major resources such as water and electrical energy is known as a “nexus” that is a cost-effective and promising solution compared to the oil, coal, natural gas, and biomass generations. This term nexus underlies to the interconnection of water, and electrical energy which is attracting the public attention and requires strict policies and specific infrastructures for each city, in order to gain the lots of its benefits that not only results in carbon dioxide reduction but also it is cost-saving and flexible (Dalla Fontana and Boas, 2019; Bijl et al., 2018). On the other hand, climate changes play an important role in the water and food security sectors. Average global warming increment will be a threatening issue for sustainable renewable resources such as freshwater, which will directly influence on agricultural industry by the 21st century. The agricultural industry has the largest participation in freshwater usage of 75%.

Furthermore, a sharp increase in the world's population is the main culprit for water and food demand. Therefore, it should be analyzed how to select which water should be used in the agricultural industry (Misra, 2014; Wallace, 2000). By considering these noticeable issues, in addition to uncertainty issues of energy de-

mand, optimization of nexus networks get more attention in different cases, which will be discussed in the following sectors.

1.1. Literature review

According to the geographical location of Iran, which is exposed by the sun and is on the direction of coming winds from the east side of the country, taking advantage of renewable energies will have positive impacts on the power supply of the country. In (Tavana et al., 2019), some strategies are considered to enhance renewable power generation besides the water demand crisis and pollution issues. In (He et al., 2019), a new power-water system with an organic ranking cycle (ORC) is proposed, and results reveal that all of the extraction parameters were effective by taking advantage of this combined system. Furthermore, a combined water-power system alongside the ORC integrated with air-heated humidification dehumidification (HDH) desalination unit is proposed to meet the water and power demands in (He et al., 2017). Harvested energy from geothermal water is used in the humidification dehumidification (HDH) desalination unit integrated with a flashing ranking cycle, which results in the effective performance of the combined system (W.F. He et al., 2018). Electrical and water demands will be raised about 70% by 2050, which is considered as a serious challenge (Shahzad et al., 2017).

Results obtained from investigating the datasheets related to the South Africa region indicate that firms without their gen-

* Corresponding authors.

E-mail address: sayyad.nojavan@ubonab.ac.ir (S. Nojavan).

Indices

g	Index of thermal generators
t	Index of time (hour)
c	Index of CWP producers
w	Index of water desalination units

Variables

$U_{c,t}$	Binary variable defined for operation status of CWP producers
$U_{g,t}$	Binary variable defined for operation status of thermal generators
$U_{w,t}$	Binary variable defined for operation status of desalination units
$P_{g,t}$	Produced power of thermal units [MW]
$W_{w,t}$	Generated pure water of desalination units [m^3/h]
$W_{c,t}$	Generated pure water of CWP units [m^3/h]
$P_{c,t}$	CWP units' production fuel cost [\$]
ζ	Robustness function of IGDT modeling

Parameters

C_p	Thermal generators fuel cost for electricity generation [\$]
C_w	Water desalination fuel cost [\$]
C_{cwp}	Both water and power producers (CWP) fuel cost [\$]
a_g, b_g, c_g	Thermal power generators' fuel cost coefficients
a_w, b_w, c_w	Water desalination units' fuel cost coefficients
$A_c, B_c, C_c, D_c, E_c, F_c$	Coefficients of CWP units production fuel cost
E_t	Power consumption [MW]
W_t	Water consumption [m^3/h]
$p_g^{\min}, p_g^{\max}$	The minimum and maximum generating power by thermal units [MW]
$p_c^{\min}, p_c^{\max}$	The minimum and maximum generated power by CWP units [MW]
$W_w^{\min}, W_w^{\max}$	The minimum and maximum generated pure water by desalination units [m^3/h]
$W_c^{\min}, W_c^{\max}$	The minimum and maximum generated water of CWP units [m^3/h]
$R_c^{\min}, R_c^{\max}$	The minimum and maximum power-water ratio of CWP unit [MWh/m^3]
UR_g, DR_g	Up and down ramp limits of thermal units
γ	Vector of input parameters' uncertainty
$\bar{\gamma}$	maximum allowable deviation of uncertain parameters from nominal values
Λ_c	New increased objective values

erator have unreliable electricity supply (Cole et al., 2018). In (F. Jabari et al., 2019), water desalination, heat, and power units are considered for a day-ahead optimization program by using demand response programs. Graph-theory-based network is considered to optimize the water-energy nexus units by using a specified and systematic process; it leads to a minimum or maximum generation in order to satisfy the external demands (Tsolas et al., 2018). In (Nojavan and Zare, 2018), a real-time pricing approach is proposed for modeling the uncertainty level by considering deterministic bi-level and epsilon constraint methods, which results in the increasing profit of the retailer. Day-ahead electricity market price clearing through the predicting methods are investigated in (Conejo et al., 2005). In (Alipour et al., 2017), stochastic programming is used for electric vehicles scheduling under market prices and the status of EVs uncertainties. Also, these electric vehicles participated under the vehicle to grid term for profit-maximizing.

Environmental issues and minimizing the total cost of generating electricity is proposed by using biogas cooling system, freshwater, and power units in (F. Jabari et al., 2019), which results indicated that considering demand response programs were effective in total electricity cost reduction. Two novel demand response programs which are depend on the household devices' uncertainty, are introduced to forecast the hourly loads in (Nakabi and Toivanen, 2019).

Two approaches, namely element nexus; which extracted water used for electricity generation and pathway nexus; which depends on the control situation and their interconnections, are proposed for wind power generation system in (Yang and Chen, 2016). A hybrid grid based on solar energy is proposed to meet the required power for a water desalination unit. Furthermore, results express that the total power generation of solar units completely satisfies the AC load of the desalination plant (Ghenai et al., 2018). In (Li et al., 2018), the concept of taking advantage of ocean energy to generate power or thermal energy is discussed to desalinate the water. Also in (Hicks et al., 1989), multilateral concepts of designing, operating basics, and results of implementing device DELBOUY are investigated. In (Hosseini et al., 2011), both equipment reliability that uses the Markov method and energy management procedure called Total Revenue Requirement (TRR) is considered for economic analysis of desalination plants, which helps plant expenses to decrease. A Game-based investigation of the energy-water nexus is also carried out in (L. He et al., 2018). Inside the life-cycle shale gas supply chain processes, a three-level paradigm for addressing the tradeoffs between energy, water, and air-emission consequences has been developed in (L. He et al., 2018). Also, the optimal and effective management of the water resources is investigated in (Cheng et al., 2016). The multi-criteria design scheme of the gas-water nexus is also proposed in (Chen et al., 2018). In (Medina-Santana et al., 2020; Fouladi and Al-Ansari, 2021; Fouladi et al., 2021), the optimal design, energy management, and integration optimization of the water-energy-food nexus is investigated.

The water's uncertainties and the power demands have a substantial impact on the nexus system's optimal operation. Ref (Guieysse et al., 2013) investigates the consequences of water uncertainty. Furthermore, ref (Ogbolumani and Nwulu, 2021) offers a stochastic fuzzy-based technique to model uncertainty in nexus systems, and (Namany et al., 2018) presented a non-cooperative game approach for uncertainty modeling. Furthermore, in (Tan et al., 2021), a stochastic optimization problem is applied for uncertainty handling in the water/power planning problem. Furthermore, (Tiwari and Adamowski, 2013) aims to study water demand uncertainty methodologies' impact and proposes a neural network-based model for forecasting.

1.2. Novelty and contribution

Uncertainty in nexus studies is one of the important challenges. As mentioned in the literature review section, there are various approaches to deal with the uncertainty of these systems. In this paper, water and power demands' uncertainties are investigated simultaneously. To solve this uncertainty, information gap decision theory (IGDT) is applied to the proposed problem. As we know, the IGDT has not been used to uncertainty modeling in the nexus systems. Besides, decision-making strategies are considered for analyzing the proposed problem. Therefore, the novelty of this paper is categorized as follows:

- Considering uncertainty for both water and power demands in the proposed water-energy nexus system
- Using IGDT method for modeling the uncertainty
- Analyzing two decision-making strategies risk-neutral and risk-averse for the proposed system

1.3. Organization of the paper

The rest of the paper is organized as follows: the considered risk-neutral and risk-averse are formulated in Sections 2 and 3, respectively, which also deals with the general conditions of applying the IGDT method for any case. Section 4 is considered to present a case study. Results are expressed in the 5th section. Finally, the last section has concluded the ideas and novelties of the proposed paper.

2. Risk-neutral based problem formulation

In this section, the deterministic formulation of the proposed nexus system is represented. The base system and related formulation are given in ref (Santhosh et al., 2014). The objective function constructed from the fuel cost of total units is minimized by Eq. (1). This cost minimization term includes seawater desalination, electrical power, and combined water and power (CWP) units. This risk-neutral term is the deterministic form of the base problem.

$$\text{Objective Function} = C_p + C_w + C_{cwp} \quad (1)$$

Eq. (2) is expressed to generate electrical power, which is a quadratic function of the thermal unit ($P_{g,t}$).

$$C_p = \sum_{g,t} (a_g P_{g,t}^2 + b_g P_{g,t} + c_g U_{g,t}) \quad ; \quad \forall g \in \Omega_p, \forall t \in T \quad (2)$$

Similar to the above equation, water desalination units' fuel cost is indicated as Eq. (3).

$$C_w = \sum_{w,t} (a_w W_{w,t}^2 + b_w W_{w,t} + c_w U_{w,t}) \quad ; \quad \forall w \in \Omega_w, \forall t \in T \quad (3)$$

Eq. (4) is formulated to address the CWP units' production fuel cost.

$$C_{cwp} = \sum_{c,t} (A_c P_{c,t}^2 + B_c P_{c,t} W_{c,t} + C_c W_{c,t}^2 + D_c P_{c,t} + E_c W_{c,t} + F_c U_{c,t}); \quad \forall c \in \Omega_{cwp}, \forall t \in T \quad (4)$$

Objective function minimization should be considered alongside electrical power, water desalination, and CWP units balance constraints. Electrical power generation and its load demand balance constraint are expressed in Eq. (5) which includes all the producing units to satisfy the electrical load demands:

$$\sum_{g \in \Omega_p} P_{g,t} + \sum_{c \in \Omega_{cwp}} P_{c,t} = E_t \quad ; \quad \forall t = 1, 2, 3, \dots, T \quad (5)$$

Similar to the electrical demand, all the pure water producing units should meet the pure water demand under the following constraint expressed as Eq. (6):

$$\sum_{c \in \Omega_{cwp}} W_{c,t} + \sum_{w \in \Omega_w} W_{w,t} = W_t \quad ; \quad \forall t = 1, 2, 3, \dots, T \quad (6)$$

By now, all the boundaries of the electrical and water desalination will be expressed. Thermal power plants and CWP units' limitations are indicated through equations (7) - (8).

$$P_g^{\min} \times U_{g,t} \leq P_{g,t} \leq P_g^{\max} \times U_{g,t} \quad ; \quad \forall g \in \Omega_p, \forall t \in T \quad (7)$$

$$P_c^{\min} \times U_{c,t} \leq P_{c,t} \leq P_c^{\max} \times U_{c,t} \quad ; \quad \forall c \in \Omega_{cwp}, \forall t \in T \quad (8)$$

Also, for pure water, equations (9) - (10) limited the water production by water desalination and CWP units.

$$W_w^{\min} \times U_{w,t} \leq W_{w,t} \leq W_w^{\max} \times U_{w,t} \quad ; \quad \forall w \in \Omega_w, \forall t \in T \quad (9)$$

$$W_c^{\min} \times U_{c,t} \leq W_{c,t} \leq W_c^{\max} \times U_{c,t} \quad ; \quad \forall c \in \Omega_{cwp}, \forall t \in T \quad (10)$$

The concept of using binary variables is to perform the on/off situations for all the producing units such as thermal generators, CWP units, and water desalination plants. Using binary variables is done to prevent the simultaneous production of the generating units. Therefore, these binary variables will be solved by mixed-integer nonlinear programming problems.

Eq. (11) will bound the water and electrical power production in CWP units.

$$R_c^{\min} \leq \frac{P_{c,t}}{W_{c,t}} \leq R_c^{\max} \quad ; \quad \forall g \in \Omega_p, \forall t \in T \quad (11)$$

Moreover, for the last constraint, ramp up and down values of thermal power plants are expressed through Eq. (12), in which the production of these thermal power plants depends on the ramp up and ramp down values.

$$\text{Max}(P_g^{\min}, P_g^{t-1} - DR_g) \leq P_{g,t} \leq \text{Min}(P_g^{\max}, P_g^{t-1} + UR_g); \quad \forall g \in \Omega_p, \forall t \in T \quad (12)$$

3. Risk-averse problem formulation via applied igdt

Different methods, including stochastic optimization, robust optimization, and information gap decision theory, are used in the literature for uncertainty modeling in optimization problems (R. Nourollahi et al., 2021; R. Nourollahi et al., 2021). In this paper, to consider the uncertainty parameters of the proposed model, according to (Nojavan et al., 2015), an IGDT optimization method is proposed. This method does not need any probability functions. The advantage of this method compared to stochastic optimization is that there is no need for uncertain parameter's historical data in the IGDT modeling. The advantage of the IGDT compared to the robust optimization is related to the ability to model the worst and good realizations of uncertain parameters based on the operator's strategy. However, the disadvantage is also related to not using the historical data in the uncertainty modeling that would be led to more conservative results. This section will analyze the procedure of the considered method, which is efficient and accurate. The general form of this approach is discussed and explained as the following equations through (13) - (20) to clarify IGDT optimization:

$$\min_X f(X, \gamma) \quad (13)$$

$$H_i(X, \gamma) = 0, \quad i \in \varphi_{eq} \quad (14)$$

$$G_j(X, \gamma) \leq 0, \quad j \in \varphi_{ineq} \quad (15)$$

$$\gamma \in \Gamma. \quad (16)$$

The vector γ is dedicated to express the input parameters' uncertainty. The term Γ is defined to express uncertain input parameters behavior. Also, X demonstrates decision variables. Eq. (17) is defined for uncertainty set explanation as following:

$$\forall \gamma \in \Gamma(\bar{\gamma}, \zeta) = \left\{ \gamma : \left| \frac{\gamma - \bar{\gamma}}{\bar{\gamma}} \right| \leq \zeta \right\} \quad (17)$$

Where the predicted value of the uncertain parameter is $\bar{\gamma}$. Also, ζ is the maximum allowable deviation of the original realization of the uncertain parameter from its forecasted value; this term is also called the radius of uncertainty, traditionally. Following Eqs. (18) - (20) will be used to bound the uncertain parameter not to deviate from its forecasted range:

$$f_b = \min_X f(X, \bar{\gamma}) \quad (18)$$

$$H_i(X, \bar{\gamma}) = 0, \quad i \in \lambda_{eq} \quad (19)$$

$$G_j(X, \bar{\gamma}) \leq 0, \quad j \in \lambda_{ineq} \quad (20)$$

In the mentioned equations, the term f_b is the resultant value of the objective function. To point out the uncertainty, the risk-averse strategy can be applied by the decision-maker with the risk-neutral strategy, which is defined as below.

3.1. Risk-neutral strategy

This strategy is the result of considering the base form of the proposed model without considering the uncertainty via IGDT.

3.2. Risk-averse strategy

This procedure uses the robustness approach for the resulting objective function (f_b) versus the worst condition, which might be occurred in forecasting the uncertain input parameter. Therefore, the decision variable set should be found optimally to maintain the actual objective function in a safe region due to the deviation of the uncertain parameter γ from its forecasted value $\bar{\gamma}$. This approach is the best option for conservative decision-makers. On the other hand, robust decisions will be satisfied in the case that the objective function is maintained against the maximum radius of uncertainty (ζ). This procedure is expressed mathematically through Eqs. (21) – (24) as follows:

$$\hat{\zeta} = \max_{\zeta, X} \zeta \quad (21)$$

$$H_i(X, \gamma) = 0, \quad i \in \lambda_{eq} \quad (22)$$

$$G_j(X, \gamma) \leq 0, \quad j \in \lambda_{ineq} \quad (23)$$

$$\left\{ \begin{array}{l} f(X, \gamma) \leq \Lambda_c \\ \Lambda_c = f_b(X, \gamma) + \xi_c |f_b(X, \gamma)|, \gamma \in \Gamma \end{array} \right\} \quad (24)$$

In the above equations, Λ_c is dedicated to a significant value that objective function should be less than it and covered in a safe region against the worst condition of the uncertain parameters. This term depends on the necessity of the decision-maker. Considering the unexpected uncertainties ξ_c is defined to specify the allowable boundary increment of the objective function (f_b). Mentioned equations have the bi-level form. For a considered X value, a maximum radius of uncertainty ($\hat{\zeta}$) is defined at the lower level (24). Now this specified value X will be used in the upper-level Eqs. (21) – (23). Decision-maker uses this value X to increase the safety region $\hat{\zeta}$. To this end, a productive and successful objective function with a specific limit is obtainable.

From now on, this mentioned risk-averse procedure will be applied to the proposed model of the paper. The following Eq. (25) will be used as a robustness function based on the IGDT approach in order to take advantage of the robustness approach.

$$\hat{\zeta} = \max_{\zeta, X} \zeta \quad (25)$$

Subject to:

Constraints (2) – (12) (26)

$$C_p + C_w + C_{cwp} \leq \Lambda_c \quad (27)$$

$$\Lambda_c = (C_p + C_w + C_{cwp}) + \xi_c (C_p + C_w + C_{cwp}) \quad (28)$$

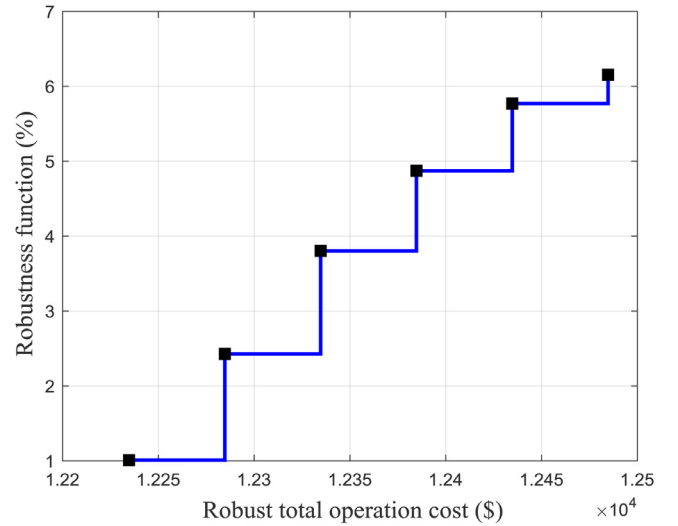


Fig. 1. Robustness function against the total robustness cost.

$$W_t = \overline{W_t} + \zeta \overline{W_t} \quad (29)$$

$$E_t = \overline{E_t} + \zeta \overline{E_t} \quad (30)$$

Besides the robustness function (25), all the proposed constraints for the base problem of the proposed paper, equations (2) – (12), will be considered to solve the problem by using the IGDT method. Also, the base objective function is limited Λ_c according to Eq. (27). The maximum deviation of uncertainty parameters formulated in Eq. (28). It should be noted that the power demand and water demand in Eqs. (5) and (6) should be updated to the new forms of Eqs. (29) and (30) in order to participate ζ in the new problem format.

4. Case study

This research studies an extensive water-power generation system constructed from four thermal power plants, three combined water power (CWP) units, and a desalination plant. All the input data of the proposed study are derived from (Santhosh et al., 2014) in detail. This considered model is a mixed-integer nonlinear program (MINLP), which is solved by the GAMS optimization software under the SBB solve that is a commercial solver for solving the MINLP problems (GAMS 2008). The global optimality of solutions under the SBB solver is ensured for convex MINLPs (Bussieck and Vigerske, 2011). Two cases are studied to clarify the effectiveness of the integration of the water-power system and to reveal the results of using IGDT optimization, as follows:

Case1. Without considering IGDT as a risk-neutral strategy

Case2. Considering IGDT as a risk-averse strategy

The represented case studies above are selected to show the results of the worst realization in the uncertain parameters. In other words, some operators may conservatively operate under an uncertain environment. The IGDT models the same condition for conservative operators.

5. Results

The following figures indicate the obtained results from optimizing the objective function by considering risk-averse and risk-neutral strategies. According to Fig. 1, the more value of robustness function increases (ζ), which indicates the worse condition of uncertain parameters; the total operation cost will increase as the exponential function. These achieved values are the same as the

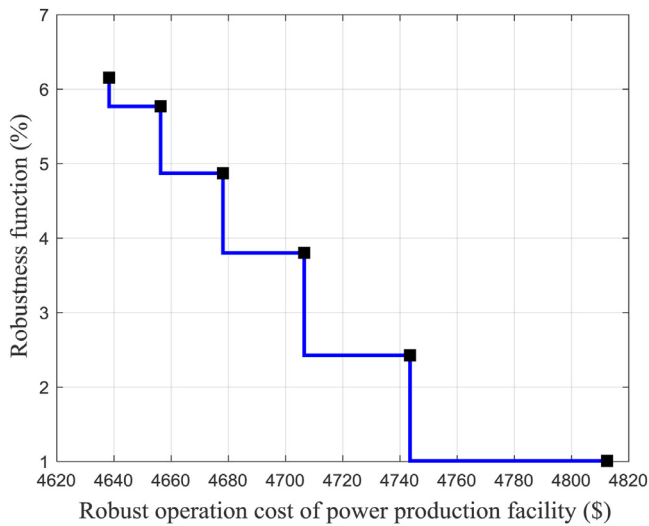


Fig. 2. Robustness function against the robustness cost of power production.

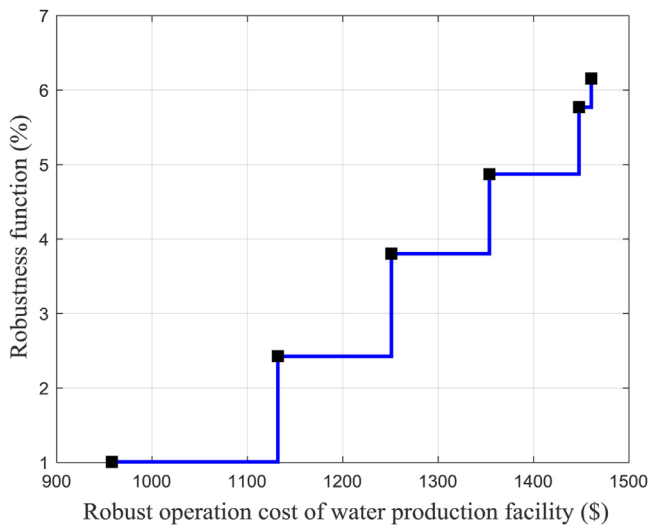


Fig. 3. Robustness function against the robustness cost of water production.

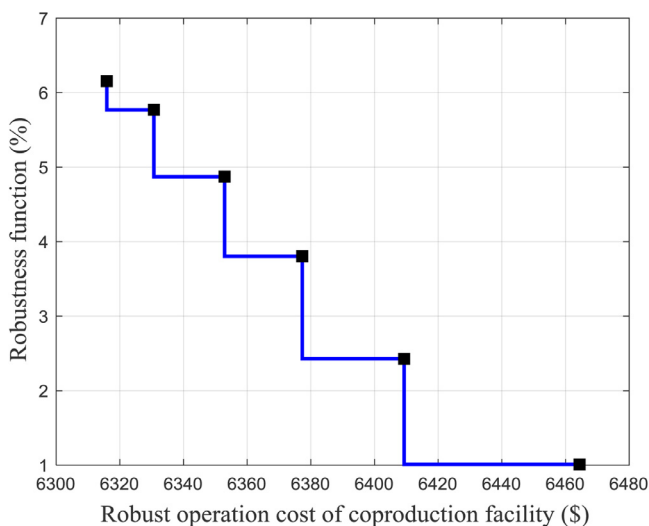


Fig. 4. Robustness function against the robustness cost of CWP production.

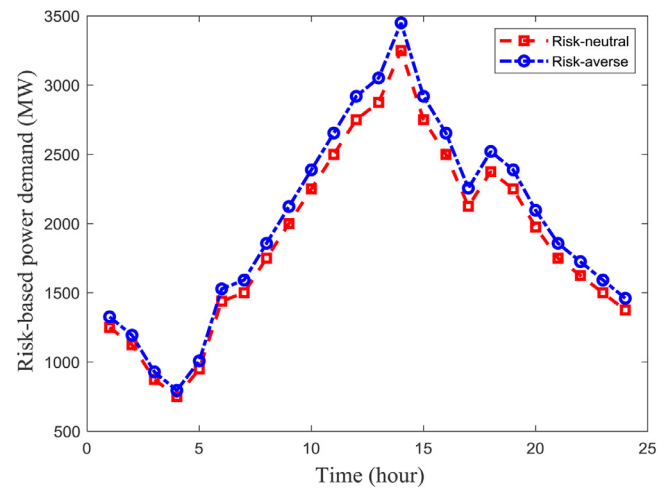


Fig. 5. Both risk-averse and risk-neutral for power demand.

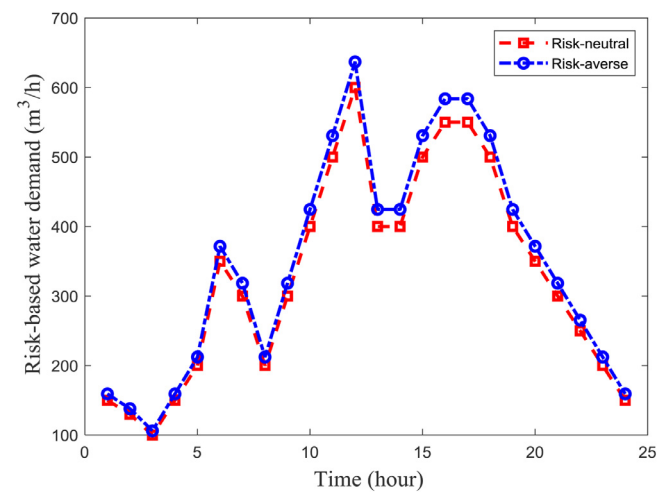


Fig. 6. Both risk-averse and risk-neutral for water demand.

expected results. More accurately, $\alpha = \zeta$ the total cost of operation is \$12,234.7. This value for $\zeta = 6.1$ which indicates that uncertain parameters have a 6.1% deviation from their forecasted value, is equal to \$12,484.7. Therefore, the total cost will increase by 2% for a 6.1% increment in the deviation of the uncertain parameters. Accordingly, after several iterations for the robustness function, it is evident that total robustness cost increment is not tangible.

In the following, obtained robustness cost of power, water, and combined water-power productions are presented to compare from its effectiveness point of view. From Fig. 2, by increasing robustness function (ζ) to a specific value, the power production cost decreases proportionally. However, at the value of $\zeta = 5.7$, the cost of power production starts to increase. Also, for ζ increment from $\zeta = 0$ to $\zeta = 5.7$, the robustness cost for power production is decreased by \$134.1, which indicates a 2.8% decrease in the cost of power production. Therefore, forecasting power demand in different cases will significantly impact the cost of power production. Not Also, lower deviations of ζ results in a sharp decrease in power production cost, but also higher deviations of ζ lead to a slow reduction one.

According to Fig. 3, which is exhibited for water production cost, it is evident that as ζ increases, the robustness cost of water production increases to a specific value of ζ . Although at the value $\zeta = 5.7$, the rate of water production cost increment is rising. Also, for an increment of ζ from $\zeta = 0$ to $\zeta = 5.7$, the robustness cost

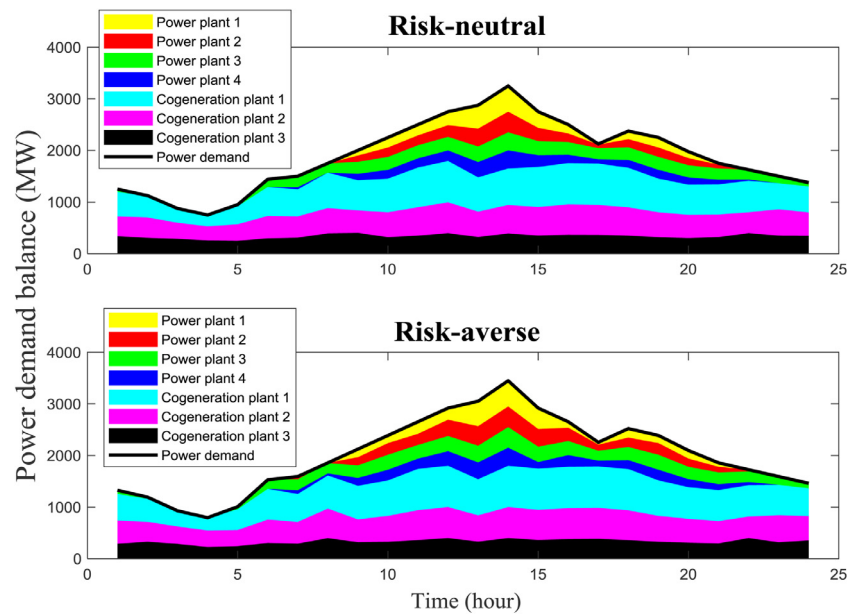


Fig. 7. Both risk-averse and risk-neutral for power demand balance.

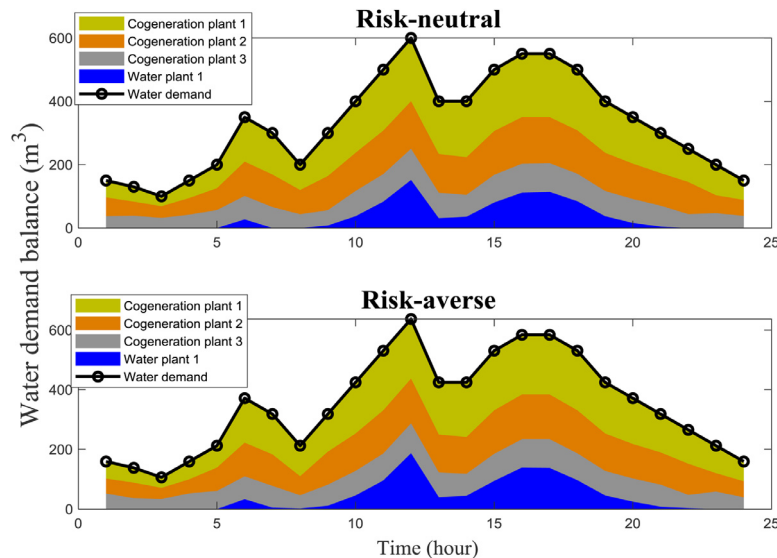


Fig. 8. Both risk-averse and risk-neutral for water demand balance.

for water production is increased by about \$502.6, which indicates a 52% increase in the cost of water production. Compared to the total cost of operation, whose curve was exponential, this function for the water production cost is linear to some extent.

From Fig. 4, similar to the cost of the power production curve exhibited above, by increasing ζ up to a particular value, the robustness cost of CWP production decreases and starts to increase. For increment of ζ from $\zeta = 0$ to $\zeta = 5.7$, the robustness cost for CWP production is decreased by \$118.5, which stands for a 1.8% decrease in the cost of CWP production. More accurately, the cost of the CWP production curve behaves similarly to the power production one, which can be concluded that power demand plays a vital role in this system, and it is the dominant curve.

Power demand based on decision-making strategies is depicted in Fig. 5 to compare the results of applying the IGDT. According to this figure, it can be observed that the difference between both

risk-averse and risk-neutral power demands is less in the off-peak hours, but this difference has been increased in peak hours, which cannot be neglected.

Fig. 6 illustrates the risk-averse and risk-neutral for water demand. Similar to Fig. 6, the difference between applied risk-averse and risk-neutral strategies for water demand is lower off-peak hours. On the contrary, the difference is noticeable in the peak hours. As a result, risk-averse water demand is higher than the risk-neutral demand.

Furthermore, Fig. 7 presented for risk-neutral and risk-averse strategies for power demand balance. All power production units meet the power demand in each hour, as exhibited by a thin black line (power demand) on the top of all production plant blocks. This satisfaction stands for a spotless implementation of the proposed model. By the arrival of peak hours, power plants 3, 4, 1, and 2 forced to operate, respectively. Also, by applying a risk-averse strategy for the proposed model, it can be seen that power plant 1 and

co-generation plant 1 increased their power production, and therefore meet the power demand, especially in peak hours in comparison to the risk-neutral strategy. Also, in some off-peak hours, power plant 2 started to generate power in the risk-averse strategy due to the pertaining condition of the considered power plant. The power plant 4 production is almost the same in both risk-neutral and risk-averse strategies. More analysis can be adopted to gain more description. As a remarkable result, it is evident that by applying the IGDT method, specifically the risk-averse strategy, the more power demand will be satisfied by the more power production in the peak hours.

Finally, similarly to the power demand balance, according to Fig. 8, water production units fulfilled the water demand as indicated by a thin black line (water demand) on the top of all water production blocks. Regarding the applied IGDT, it can be understood that water plant 1 production is increased its production in the risk-averse strategy with a significant difference in the peak hours. Also, in both decision-making strategies, water plant 1 is used to operate during peak hours. More analysis can be investigated to obtain more results. Therefore, this applied IGDT method's importance cannot be ignored because it has a remarkable impact on power and water production.

6. Conclusion

In this paper, the provided energy-water nexus model is robustly scheduled using IGDT. The risk-averse operation of the studied nexus system includes water, power, and CWP units. In conclusion, the uncertainties of the different considered units have remarkable impacts on the total operation cost of the studied nexus model. In this study to analyze the studied test system, two case studies named risk-averse strategy and risk-neutral strategy are proposed. According to the results, in the risk-averse case, both power and water demands are increased, and water plant 1, and power plants 1, 2, 3, and 4 are used to meet the demand at peak hours. Also, operation scheduling is studied considering the uncertainty parameter ζ . As results demonstrate, by increasing the level of a considered uncertain parameter, the total cost of the studied model is increased slightly. To be more precise $\zeta = 0$, which indicates the risk-neutral strategy, the total cost of the studied model is about \$12,234.7. Also, by increasing of ζ from 0 to 6.1, the total cost increases to \$12,484.7. The future work of this paper could be capacity planning problem of a similar nexus system.

Declaration of Competing Interest

The authors have no conflict of interest.

CRediT authorship contribution statement

Peiru Jian: Data curation, Methodology, Software, Writing – original draft. **Qun Guo:** Formal analysis, Validation, Visualization, Writing – original draft. **Sayyad Nojavan:** Conceptualization, Investigation, Supervision, Writing – review & editing.

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